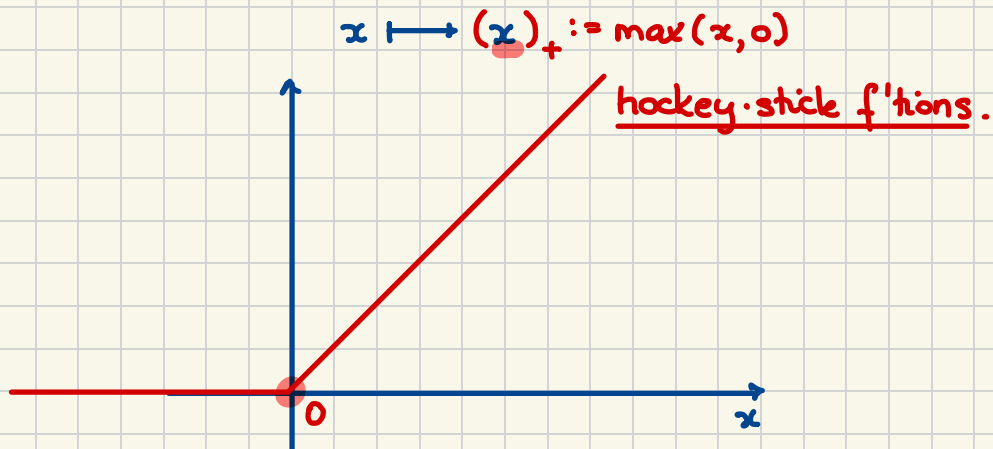
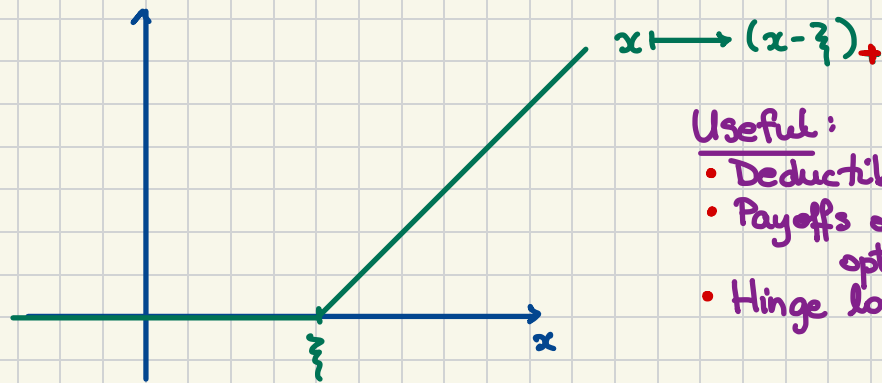


Linear Splines.

Positive Part Function.



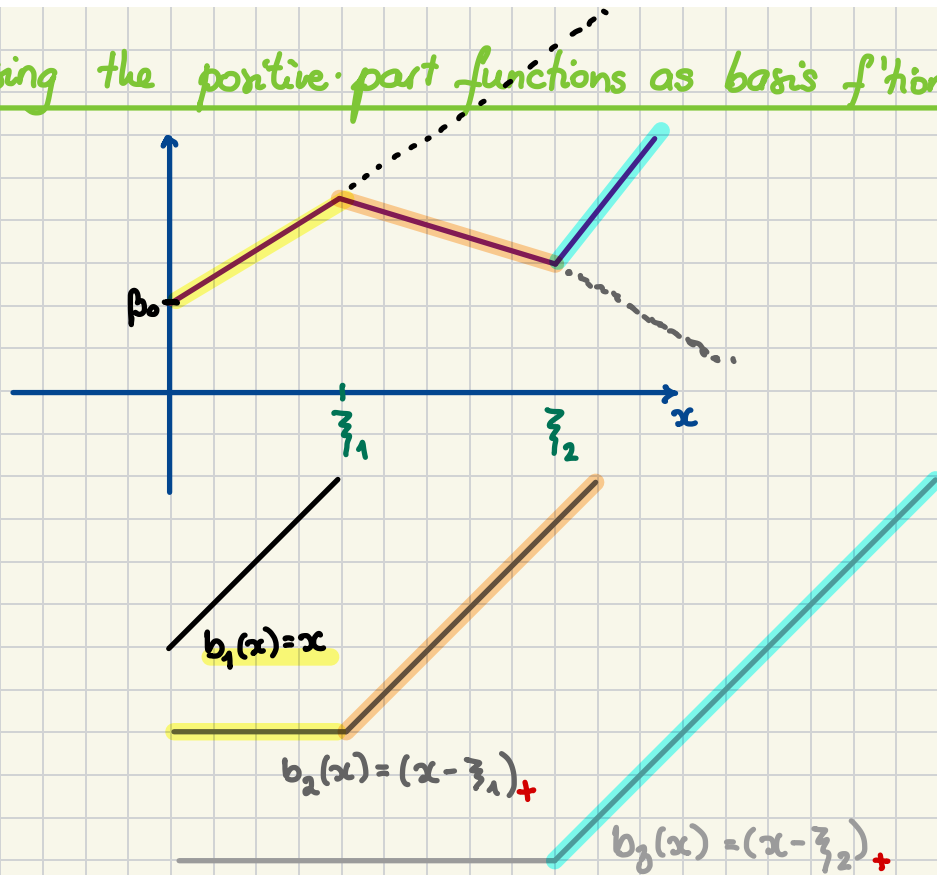
Example. For any $\xi \in \mathbb{R}$, let's graph the basis f'ion corresponding to it



Useful:

- Deductibles
- Payoffs of options
- Hinge loss

Using the positive-part functions as basis f'ctions.



$$\mathbb{E}[(g(x) - y)^2 | X=x] \xrightarrow[g]{\min}$$

$$g(x) = \beta_0 + \beta_1 b_1(x) + \beta_2 b_2(x) + \dots$$